The Futuristic History project: recreating history with augmented reality solutions

Tuomas Mäkilä, Seppo Helle, Lauri Viinikkala, Teijo Lehtonen, Juha-Pekka Arimaa,
Olli Heimo, Päivi Nikkola, Timo Korkalainen and Veijo Pönni,

University of Turku, Finland

https://www.utu.fi/fi/yksikot/bid/projektit/Sivut/Futuristic_History.aspx

Abstract: Futuristic History is a two-year research project started in January 2013, implemented in cooperation with University of Turku and VTT Technical Research Center of Finland. Its goal is to research and develop augmented reality based solutions and business models for tourism and culture. TEKES, The Finnish Funding Agency for Technology and Innovation, is funding the project and there are several partners from ICT, travel and museum sectors.

The project creates working prototypes, which present historically important locations in new, interesting ways. Special focus is placed on augmented reality techniques, which mix digital content with real-life environments. Finding economical and efficient ways to make such content is also an essential part of the work. It is investigated what kind of business models would be feasible and attractive in museum context.

The target locations of the project are in the region of Turku, which is the former capital of Finland and has several historically interesting sites. We have already completed a mobile 3D virtual guide of a short-lived 16th century church in Turku. Other target locations include Luostarinmäki Handicraft Museum, which is the only preserved part of the 18th century wooden city of Turku, and the Louhisaari Manor, one of the rare palatial style manors in Finland. In these locations, the original residents' daily life and work will be presented using augmented reality solutions. We may also be able to reveal hidden details or let physically disabled persons look through walls of difficult-to-access rooms. In central Turku, old photographs will be used to give a virtual view of some buildings that have been demolished long ago.

We present the goals and production principles behind the prototypes implemented during the project. We also briefly discuss the challenges and opportunities of bringing augmented reality solutions to museum settings from the perspectives of both museum professionals and end-users.

Keywords: augmented reality, mixed reality, 3D modeling, museums

1. Background

Augmented reality (AR) can be defined as real world environment that is augmented by computer-generated content, e.g. 3D models, images, sounds or textual information (Milgram & Kishino 1994). Augmented content is presented in real time, positioned accurately in relation to the real objects, and the user can interact with it. *Virtual reality* consists of virtual, i.e. computer-generated, content only, and a combination of virtual and augmented reality is often called *mixed reality*.

A wide variety of different AR applications exist. The simplest AR applications from the immersion perspective are so called AR browsers, which usually overlay contextual text information onto video stream through a mobile device. For example, an AR browser could show the directions to the nearest sightseeing attractions (see e.g. http://www.wikitude.com/). On the other end, there are highly immersive 3D applications, which basically surround the user with computer-generated content that is seamlessly integrated into the real-life surroundings.

In museums, AR technology can create realistic, immersive experiences that would be impossible to achieve with traditional means like printed media. Buildings and lost settings can be reconstructed virtually and historical events can be recreated. Such solutions can be used for educational purposes, and they may be able to e.g. increase the general public's interest towards historical sites and events. The use of augmented reality in the museums has been researched for several years, especially from the technological perspective, e.g. (Wojciechowski 2004; Miyashita 2008; Barry 2012), but the technology is still on the verge of wider adoption.

2. Goals of the project

Futuristic History[1] is a research project conducted by the University of Turku and the VTT Technical Research Centre of Finland. The general goal of the project is to investigate the use of augmented reality technologies in museums and other historical sites. The project is funded by TEKES, the Finnish Funding Agency for Technology and Innovation, as a strategic research; this means that a roadmap for continuing the national research on the topic will also be formed during the project. The project will last from January 2013 to December 2014.

The project analyzes the current state of augmented reality in museum applications by developing augmented and mixed reality applications for tourism and museums. It is a multidisciplinary undertaking where several different aspects of the content production are being researched.

Content production is done in cooperation between people who know the history of the target location and those who can make an attractive application of it, such as writers, designers, software and marketing professionals. The project aims to find efficient ways for different professionals to cooperate. It is also important to find business models that are profitable. The new solutions should be attractive enough to increase the amount of visitors and produce more income to museums.

Ultimately, the project results should: 1) lead to more funding for the operating of historical sites, 2) encourage more people to visit historical sites, and 3) increase public awareness of historical knowledge and facts. Other local businesses may also benefit from the increased interest of a town or site. Finally, there is an opportunity for new business in the production of applications and related tools.

3. Prototype implementations

The project will implement prototype applications concentrating on a few sites. These were selected based on general interest and potential for mixed reality applications. We also paid attention to having diverse sites to produce a varying collection of applications. The sites vary in their historical background and setting, and in the characteristics of augmented reality application.

Holy Ghost Church: The first implementation done in the project was a 3D model of the Holy Ghost Church in Turku and an application for virtual observation of the model. The construction of the original church started in 1588 and it was intended for the use of the Finnish-speaking citizens of Turku; there was already a cathedral in the town. The new church was damaged in a fire in 1593, and afterwards the ruins of the church were used as a cemetery until the church was demolished in the 1650s. The application produced, presented in Image 1, offers a virtual reality tour of the interiors of the church. The model and its production process are presented in more detail by Viinikkala (2013).

Luostarinmäki Handicrafts Museum: This handicrafts museum is in the only preserved part of the 18th century wooden city of Turku, as illustrated in Image 2. It consists of a number of houses and yards along a few narrow streets. The houses are furnished, so there is plenty of content available. For this location we are creating a game in which the visitor will be doing tasks to help with wedding arrangements. It will require searching for places and objects as well as communicating with people. Most of the game characters are digital, but visitors can also communicate with the guides and craftsmen present at the museum. The environment offers a lot of possibilities for the application of virtual content in the real environment. One goal is to record the work of the craftsmen, as, in some cases, very few people remain who know the old skills.

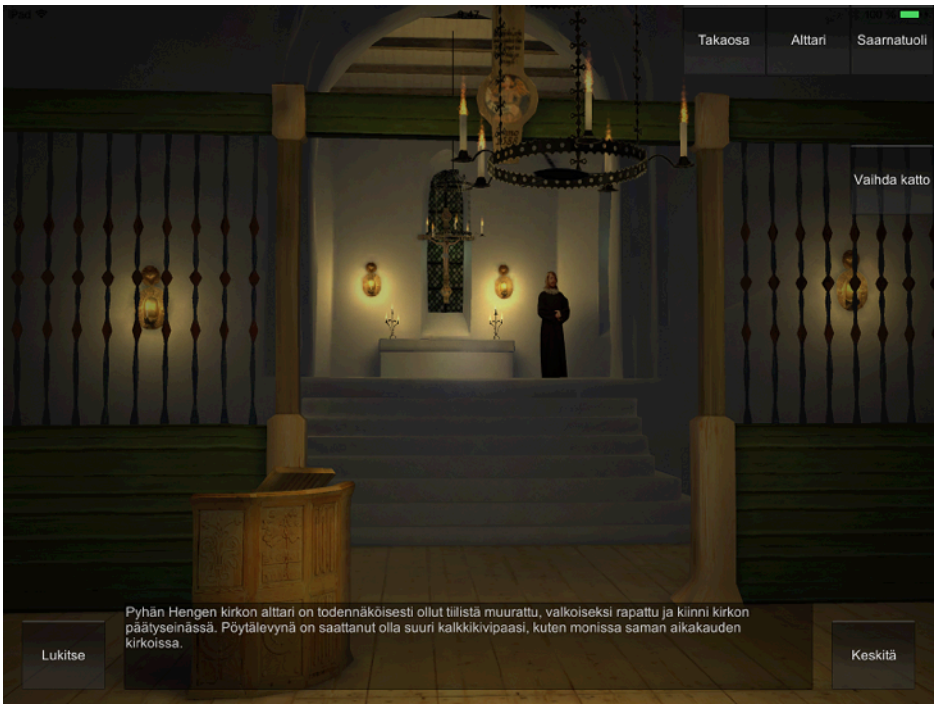
Louhisaari Manor: Finland has very few palatial style manors; Louhisaari, situated circa 20 km northwest of Turku, is one of them. While the estate had been belonging to the Fleming family from the 15th century, the palatial main building, presented in Image 3, dates from 1655. Visitors can enter the main building on guided tours only, which means that an application for individual use inside the main building is not the ideal approach. Therefore the focus is on the surrounding garden and other buildings. The new main building was originally at the seashore, but due to still continuing rising of the ground after the ice age, the shoreline is now several hundred meters away. The change of the view from the manor, the development of the gardens and the changes in surrounding buildings will be presented in the application that is being planned for Louhisaari. The daily life of servants and other local people will also be presented around the surroundings.

Central Turku: Turku is the oldest town in Finland, and it was the capital of the country until 1812. In 1827 fire destroyed most of the town center, which consisted mainly of wooden buildings. After that,

[1] Futuristic History: user-centric mixed reality applications for presenting, recreating and restoring historical events and place.

the plan of the town center was renewed and wooden buildings were not allowed along main streets. Many of the 19th century buildings have, however, been demolished during the last decades of the 20th century. One application in the project will present some of these buildings, as seen in photographs from the19th and early 20th centuries, using augmented reality solutions.

Image 1: Holy Ghost Church Virtual Reality Application



Screenshot of the application running on iPad. The priest standing in the altar is a 2D video embedded into the 3D model. Real Finnish chant of the era is used as background music.

Image 2: Luostarinmäki Handicrafts Museum



Photograph of a street in the museum. In the pilot application the existing surroundings are augmented mainly with 3D models of 19th century people and domestic animals to illustrate the museum area as a city quarter.
Image 3: Louhisaari Manor



Photograph of the main building of the manor. The manor has beautiful garden areas, which have changed significantly during the centuries. One way to utilize AR technologies is to illustrate this change.

4. Challenges and opportunities of augmented reality in museums

Augmented and mixed reality have the potential to deliver attractive experiences and to raise interest about historical sites among the general public. Virtual content can offer realistic experiences of the events and objects of history. However, creating those experiences requires a lot of work and can easily fail.

Quality of the content: The content delivered to end-users must be high quality and attractive, but also affordable. At the same time, it should offer accurate information, especially about historical content. There can be fictive elements, but facts should not be distorted. Content creation is a cooperative process between collaborators, e.g. history experts on the target site, writers, software developers and game designers.

Technical feasibility: The content must run in the available hardware and be easy to use. Depending on the delivery model, support for many different devices may be required, e.g. when visitors use their own devices. Devices with the least performance capacity may set the baseline for available content quality in a technical sense, while higher performance devices could do heavier processing and support more complex features. Building different versions of the application for different end-user devices would increase the complexity and the price of the content creation process.

Usability: Tourism-related context implies that a large portion of users may use the particular application first time. Thus, the usability aspects become important: the application should be easy to learn. This problem may be decreased with a common platform on top of which different content applications are made. So, the user might already know the common ways of interaction even though the content for the particular location would be new.

Delivery model: How to deliver the content to the public? Lots of people already have suitable equipment - smartphones and tablets - so an approach is to offer content to be downloaded to the user's device. This, however, causes the need to support a potentially large variety of different devices, and delivering the - possibly large - application to the device is also a technical issue.

Another approach is to loan/rent equipment for the visitors at the site. This would guarantee the compatibility and performance of the hardware as well as solve the question of software delivery. On the other hand, a pool of rented devices is an additional cost, and in this approach the users can't use the application after they have returned the device. It may be a significant part of the business model to offer some content that is available before the visitor comes to a museum and after the visitor returns from the museum.

Content production process: Perhaps the major work in creating applications is the design and technical input of the actual content. Multidisciplinary augmented reality professionals are needed to produce quality content cost-effectively. However, the produced content is seldom static and therefore the local museum personnel most probably have to have basic tools and sufficient skills to update the content independently.

Profitability and business models: Augmented reality applications are specifically related to a certain geographical location. Therefore, especially in smaller museums, the production and the maintenance cost have to be optimized and the income streams and models carefully planned. Different revenue models, e.g. paid applications, freemium[2] models and application costs included in the ticket prices, all have advantages and challenges. Finding working business models for augmented reality applications is essential for the wider adoption of AR technologies in museums.

5. Conclusions

The use of augmented reality in museum contexts has been researched but there are still many unsolved questions relating to e.g. technological challenges, content production processes and feasible business models. The Futuristic History project tries to solve some of these challenges and to pinpoint the most important ones for future research. This is achieved through several prototype implementations of augmented reality applications in two different museums, in a historical site and at the center of the historical city of Turku.

Acknowledgements

The project is funded by Tekes, the Finnish Funding Agency for Technology and Innovation. We appreciate also the partnership of public and private parties: the National Board of Antiquity, the Finnish Tourist Board, Lounaispaikka, Metaverstas Ltd, Serious Games Finland Ltd, Nokia Research Center, Turku Touring, Muuritutkimus and Casagrandentalo.

References

Barry, A., Thomas, G., Debenham, P., & Trout, J. (2012). Augmented Reality in a Public Space: The Natural History Museum, London. Computer, Vol 45, Issue 7, pp 42-47, July 2012. IEEE.

Milgram, P., & Kishino, F. (1994). A taxonomy of mixed reality visual displays. IEICE Transactions on Information Systems, Vol E77-D, No.12 December 1994.

Miyashita, T., Meier, P., Tachikawa, T., Orlic, S., Eble, T., Scholz, V., Gapel, A., Gerl, O., Arnaudov, S., & Lieberknecht, S. (2008). An Augmented Reality Museum Guide. Proceedings of the 7th IEEE/ACM International Symposium on Mixed and Augmented Reality, pp 103-106, September 2008. IEEE.

Viinikkala, L., Korkalainen, T., Arimaa, J.P., Lehtonen, T., Helle, S., Mäkilä, T., Pönni, V., & Heimo O. (2013). Mixed Reality and the Holy Ghost Church in Turku. Nordic Digital Excellence In Museum Conference 2013.

Wojciechowski, R., Walczak, K., White, M., & Cellary, W. (2004). Building Virtual and Augmented Reality Museum Exhibitions. Proceedings of the ninth international conference on 3D Web technology, pp 135-144, April 2004. ACM.

[2] Freemium means that the application itself is free but there is additional content with extra cost.